Verification and Validation Report: Software Engineering

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1 Revision History

Date	Version	Notes
March 8th, 2026	1.0	Added Non-Functional Performance tests
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed —SS]

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3 Functional Requirements Evaluation

4 Nonfunctional Requirements Evaluation

4.1 test-PF-1: ASR Latency Benchmark

Type Automated, Dynamic

Test Case Derivation PRF-13.1

Test Summary The ASR latency benchmark was executed using 20 audio samples spoken by various speakers. For each sample, latency was measured from the marked end-of-speech point to the time at which the transcript became available. The measured latencies were recorded and averaged to evaluate performance against the requirement threshold.

Table 1: ASR latency benchmark results for 20 audio samples

#	**Sample Text**	**Length (words)**	**Latency (s)**
1	open google	2	5.81
2	google.com	1	6.30
3	search Wikipedia about paris	4	6.08
4	navigate to paris's sport section	5	5.93
5	open gmail	2	4.73
6	open google and search for keyboards	6	5.03
7	go on bestbuy and search for newest game	8	5.43
8	search software engineering jobs	4	4.48
9	search how to boil pasta	5	5.21
10	search used honda civic	4	6.08
11	search toronto raptors schedule	4	6.50
12	go to the history section of the canada page	9	7.01
13	search apple stock price	4	4.81
14	search directions to union station	5	5.21
15	open reddit	2	6.00
16	play jazz music on youtube	5	6.03
17	search restaurants near me	4	5.95
18	open wikipedia	2	4.83
19	search mcmaster university	3	6.80
20	open ebay	2	5.00
		Average Latency	**5.66**

Test Data and Results

Calculation of Average Latency The average latency is calculated as:

$$\text{Average Latency} = \frac{\sum_{i=1}^{20} L_i}{20} = 5.66 \text{ s}$$

where L_i is the latency measured for sample i.

Actual Result The measured average latency across the 20 audio samples was 5.66 seconds.

Conclusion This test **does not fully satisfy** PRF-13.1, since the measured average latency exceeded the required threshold of 5 seconds.

4.1.1 test-PF-2: Command Pipeline Latency

Type Automated, Dynamic

Test Case Derivation PRF-13.2

Test Summary The command pipeline latency benchmark was run using the same 20 audio samples from the ASR latency test. For each sample, latency was measured from the point at which the transcript became available to the point at which the requested command result was completed by the system. The measured latencies were recorded and averaged to assess compliance with the performance requirement.

Table 2: Command pipeline latency results for 20 audio samples

#	**Sample Text**	**Length (words)**	**Latency (s)**
1	open google	2	8.92
2	google.com	1	9.41
3	search Wikipedia about paris	4	12.87
4	navigate to paris's sport section	5	14.35
5	open gmail	2	9.76
6	open google and search for keyboards	6	13.42
7	go on bestbuy and search for newest game	8	15.91
8	search software engineering jobs	4	12.26
9	search how to boil pasta	5	11.83
10	search used honda civic	4	13.74
11	search toronto raptors schedule	4	12.95
12	go to the history section of the canada page	9	18.62
13	search apple stock price	4	10.54
14	search directions to union station	5	14.08
15	open reddit	2	9.88
16	play jazz music on youtube	5	13.67
17	search restaurants near me	4	12.43
18	open wikipedia	2	9.15
19	search mcmaster university	3	11.96
20	open ebay	2	8.64
		Average Latency	**12.52**

Test Data and Results

Calculation of Average Latency

$$\begin{aligned}\text{Average Latency} &= \frac{1}{20}(8.92 + 9.41 + 12.87 + 14.35 + 9.76 + 13.42 + 15.91 + 12.26 + 11.83 + 13.74 \\ &\quad + 12.95 + 18.62 + 10.54 + 14.08 + 9.88 + 13.67 + 12.43 + 9.15 + 11.96 + 8.64) \\ &= 12.52 \text{ s}\end{aligned}$$

Actual Result The measured average command pipeline latency across the 20 audio samples was 12.52 seconds.

Conclusion This test satisfied the performance requirement, since the measured average latency was below the 15 second threshold.

4.1.2 test-PF-3: Accuracy & Precision (Stationary Noise)

Type Offline Evaluation

Test Case Derivation PRF-13.1, PRF-13.2

Test Summary This test was planned to evaluate ASR and command classification performance under stationary background noise conditions, including AC and fan noise mixed at 20 dB, 10 dB, and 5 dB SNR. The intended evaluation metrics were word recognition accuracy and command precision under noisy conditions.

Result This test was not executed during the current development cycle.

Rationale Robustness to stationary background noise was identified as a stretch goal for the project. At this stage, development and validation efforts were prioritized toward core functionality under standard operating conditions. As a result, noise robustness testing was deferred and is outside the current project scope.

Conclusion No result is reported for this test. Evaluation of performance under stationary noise remains future work.

4.2 Usability

4.3 Performance

4.4 etc.

5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted. —SS]

8 Automated Testing

9 Trace to Requirements

10 Trace to Modules

11 Code Coverage Metrics

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?

4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)